An example mapping from some routinely offered courses to the steps of the Data Science Life Cycle.

The table is not intended as a complete and comprehensive description of all skills required to be an effective data scientist, but an illustration of how current courses could be incorporated into a data science training curriculum, within which students may pursue pathways of interest. Possible new courses to be developed can be gleaned from such a presentation. Some courses are listed in more than one step to illustrate various ways they might be included in curriculum design.

Data Science Life Cycle Step	Possible (Existing) Courses
Experimental design	▶ Introduction to Probability ▶ Introduction to Statistics ▶ Design of Experiments (including Human Subjects and Informed Consent)
Obtaining data	▶ Experimental Methodology ▶ Introduction to Databases ▶ Introduction to SQL, noSQL ▶ Sensor Integration and Control
Data exploration	▶ Introduction to R ▶ Introduction to python ▶ Graphics and Data Visualization ▶ Introduction to Statistics
Databases and data structures including cleaning/organizing	▶ Introduction to Database Systems ▶ Introduction to SQL, noSQL ▶ Natural Language Processing (NLP)
Software engineering	▶ Python, R, C, C++, Julia ▶ Distributed Systems, MapReduce ▶ Software Testing
Feature selection	▶ Statistical Learning ▶ Domain-specific courses, for example, Bioinformatics for Transcriptomics; Brain Imaging in Cognitive Neuroscience Research
Model estimation	▶ Mathematics (Probability, Linear Algebra, Calculus, Real Analysis) ▶ Applied Statistics ▶ Machine Learning ▶ Data Mining ▶ Deep Learning ▶ Scalable Algorithms ▶ Statistical Decision Theory
Simulation and cross-validation	▶ Fundamentals of Numerical Methods ▶ Introduction to Computer Modeling and Simulation ▶ Statistical Learning
Visualization	▶ Information Visualization ▶ Scientific Visualization and Graphics ▶ [Domain specific courses such as Learning ArcGIS; Spatial Data Visualization]
Publication/Archiving	▶ Introduction to Information ▶ Data Archiving and FAIR Data ▶ Scientific Report Writing ▶ Research Data Management ▶ Open Access and Scholarly Communication ▶ Digital Libraries and Preservation
Overarching topics	▶ Ethics for Scientists ▶ Data Privacy ▶ National and International Regulatory Trends in Data Protection